SHIP DESIGN METHODOLOGY

NTA LEVEL 6 & BRIDGE COURSE

SRT 06107 & SRT G6107

Lecture contents

- Estimation of steel weight for a new ship.
- Methods for estimating steel weight in ships
  - Cubic Number method.
  - Weight per meter method.
  - ‘Slog-slog’ method.
  - Method of differences.
  - Computational techniques.

Estimation of Steel Weight for A New Ship

- For every ship there is a ‘balance of weights’ table, an example of which is shown in Table 1.0. This shows the actual figures for a Shelter deck General Cargo vessel of 128 m length between perpendiculars (LBP).

Table 1.0 A balance of weights table (tonnes)

Steel weight	2800
Wood and Outfit weight	700
Hull weight	3500
Machinery weight	550
Lightweight	4050*
Deadweight	9050
Fully loaded weight	13 100

- The Naval Architect will always attempt to make the lightweight as low as possible without endangering the safety and strength of the new vessel.
- The Department of Transport (DfT) and International Maritime Organization (IMO) keep a watchful eye on the safety standards whilst Lloyds are more concerned with the strength considerations. Other countries have equivalent Classification Societies.

- **Net scantling weight**: This is the steel weight that is actually ordered in by the shipyard. It is subjected to a rolling margin of -2.5% to +2.5% of the thickness of each plate.
- **Invoice weight**: This is the steel purchased by the shipyard.
- **Net steel weight**: This is the weight that ends up in the new ship. It takes into effect the wastage caused by plate preparation. The steel that ends up on the cutting floor can be 8–10% of the delivered plate. Figure 2.1 shows a nested plate with wastage material regions.

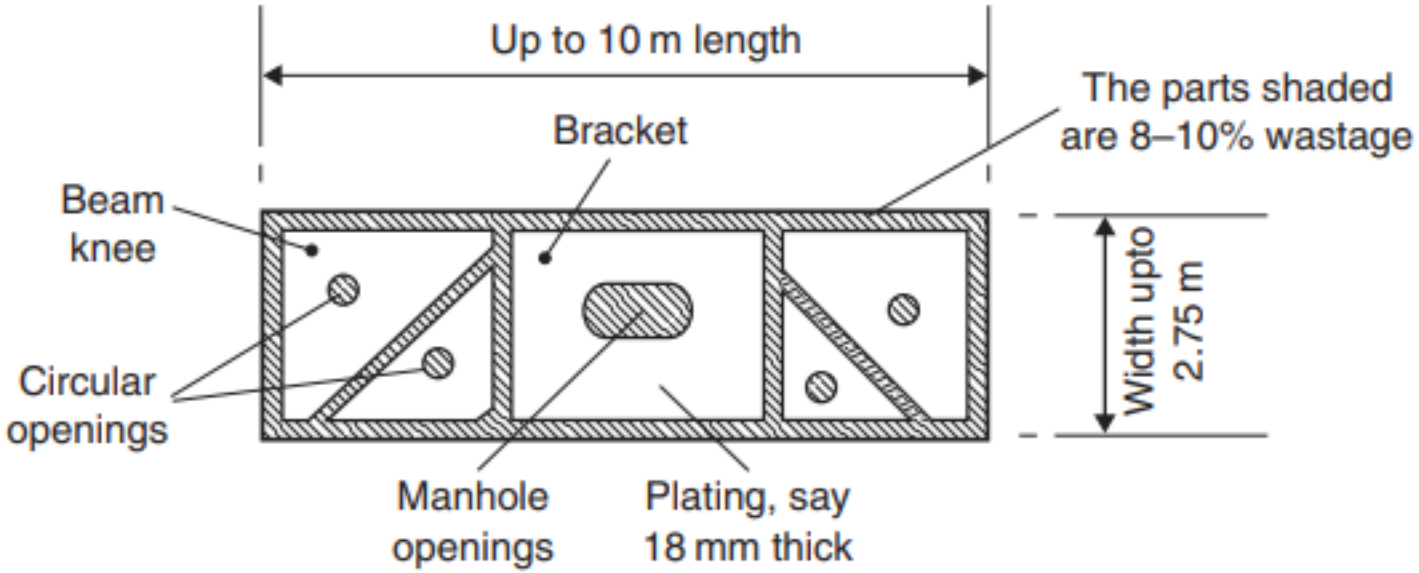


Fig. 2.1 A nested steel plate.

Methods for estimating steel weight in ships

- There are several methods for obtaining the steel weight of a new design some of them being:
  1. Cubic Number method
  2. Weight per metre method
  3. ‘Slog-slog’ method
  4. Method of differences
  5. Computational techniques

Cubic Number method

Initially, this method estimated the steel weight by using the cubic number that we obtain from the product of length, breadth and depth.

So that the steel weight = $C\ \frac{L \times B \times D}{10}$ where C is the coefficient obtained by dividing the steel weight of the basic ship by its cubic number.

A superior method is to obtain proportionate dimensions for the new design, assess the steel weight on the ratios of lengths cubed and then correct for breadth and depth and adjust for differences in structural lay out.

Worked example

A basic ship is 121.95 m × 16.46 m × 9.15 m Depth Mld with a total steel weight of Ws. A new similar design is 131.1 m × 17.07 m × 10.06 m Depth Mld. Show the method for obtaining the steel weight for the 131.1 m ship.

Worked example

	L	B	D	
Basic ship	121.95	16.46	9.15	
Basic ship × L_2/L_1	131.10	17.70	9.83	..... (1)
New Design	131.10	17.07	10.06	.....(2)
Eqn (2) – (1) =	zero	-0.63	+0.23	

Thus the steel weight for the new design = W_s × $(131.10/121.95)^3$ but with a deduction for 0.63 m of breadth and an addition of 0.23 m for depth. The fourth method later shows how the adjustments are then made for this deduction and this addition.

Weight per meter run method

- In this method it is necessary to have the midship sections of the basic ship and for the new ship.
- Calculations are made to obtain the weight per meter run at amidships for both ships. In these calculations only longitudinal plating and longitudinal stiffening are considered. Intercostal steel structures are ignored.

Worked example

Weight per metre run (W_b) for a basic ship is 13.12 tonnes/m. LBP is 118 m and steel weight is 2350 tonnes. From the preliminary midship section for the new design, the weight per metre run (W_d) is 13.76 tonnes/m, LBP is 124 m. Estimate the steel weight for the new design.

Worked example

Let steel weight for the basic ship = W_B

Let steel weight for the new design = W_D

Then $$W_D = W_B \times \left(\frac{W_d}{W_b}\right) \times \left(\frac{L_2}{L_1}\right)$$

$$= 2350 \times \frac{13.76}{13.12} \times \frac{124}{118}$$

$$= 1.102 \times 2350$$

$$= 2590 \text{ tonnes}$$

Note that this is only a first approximation and must always be treated as such. There are certain assumptions made with this method.

The ‘slog-slog’ method

- This method is used where a basic ship is not available. It requires a preliminary set of steel plans for the new design.
- Length, breadth and thickness of the steel plates and stiffeners are multiplied together, and then added to give a total volume of steel.
- Any openings in the steel have to be allowed for and deducted from this volume.

- By bringing in the specific gravity for steel of about 7.85, the volume can be changed to steel weight.
- Being very repetitive in nature it is very tedious. It can take a long time to obtain the final steel weight. This is why it is known as the ‘slog-slog’ method!!

Method of differences

- In this method, dimensional correction is made for length, breadth and depth after comparisons have been made between the new design and a selected basic ship.
- Feedback from ships already built has shown that the steel weight in tones/m run for length, breadth and depth are as follows:

- 85% is affected by length of a ship,
- 55% is affected by the breadth of a ship,
- 30% is affected by the depth of a ship,
- 45% is affected by the depth of a ship for Oil Tankers only.

The percentages take into account end curvature of vessels and curvature below say the Upper Deck level.

Worked example

A General Cargo ship is 122 m ×16.45 m × 9.20 m Depth Mld. She has a finished steel weight of 2700 tonnes. The new ship has preliminary dimensions of 131 m × 17.08 m × 10.10 m Depth Mld. Estimate the steel weight for the new design after correcting for the Main Dimensions only.

For the basic ship:

Rate along the length = 85% × (2700/122) = 18.81 tonnes/m run

Rate across the breadth = 55% × (2700/16.45)= 90.27 tonnes/m run

Rate down the depth = 30% × (2700/9.20) = 88.04 tonnes/m run

	L	B	D	
Basic ship	122	16.45	9.20	
New design	131	17.08	10.10	
Differences	+9	+0.63	+0.90	
Rates in tonnes/m run	18.81	90.27	88.04	
Modifications	+169	+57	+79	= +305 tonnes

So, new design's steel weight = basic steel weight + modifications

= 2700 + 305

= 3005 tonnes

after modifying for Main Dimensions only!!

- Note how the three rates in tones/m for the basic ship, are also used for the new design. It should also be realized that any or all of the three modifications can be positive, zero or indeed negative.

Class Activity

A basic General Cargo ship is 135 m × 18.53 m ×10.0 m Depth Mld with a finished steel weight of 3470 tonnes. A new design is 136.8 m × 18.36 m × 9.8 m Depth Mld. Estimate the steel weight for the new design after modifying for Main Dimensions only.

Computational techniques

- Many formulae have been suggested by researchers for estimating the finished steel weight.
- Three of them have been J.M. Murray, I. Buxton and S. Sato. They were derived after keying in and plotting a lot of detailed total steel weights from ships already built and in service. They were all for Supertankers.

$$W_{ST} = 26.6 \times 10^{-3} \times L^{1.65} \frac{(B + D + H/2)(0.5C_B + 0.4)}{0.8} \text{ tonnes}$$

J.M. Murray (1964)

$$W_{ST} = a_1 \times (L^{1.8} \times B^{0.6} \times D^{0.4})(0.5C_B + 0.4) \text{ tonnes}$$

I. Buxton (1964)

$$W_{ST} = \left(\frac{C_B}{0.8}\right)^{1/3} \times 10^{-5} \left\{\left(5.11 \times L^{3.3} \times \frac{B}{D}\right) + (2.56 \times L^2 \times (B + D)^2)\right\} \text{ tonnes}$$

S. Sato (1967)

where:

L = LBP,

B = Br.Mld,

D = Depth Mld

H = Draft Mld

C_B = block coefficient at SLWL

a_1 = Buxton’s coefficient to obtain units of tons (or tones).

- These formulae serve only to give only first approximations to the steel weight.
- As ship main proportions have changed over the years and as high tensile steel became more used in these Supertankers then the coefficients will also have changed with time.
- Treat these computer derived formulae with caution, and certainly only as a first guidance to the finished steel weight.

Of the five methods discussed, it is suggested the best one is the ‘Method of differences.’

Any Questions, Suggestions, Comments, etc.

THE END

